this quantity is the speed of the travelling wave B1 [3]

5	**(a)**	**(i)**	F/A	**B1**	
		(ii)	$\Delta L/L$	**B1**	
		(iii)	$FL/A.\Delta L$	**B1**	**[3]**
	(b)	**(i)**	$\Delta L = 0.012 \times 0.62 \times 350$	**M2**	
			$= 2.6$ mm	**A0**	**[2]**
		(ii)	$2.0 \times 10^{11} = (F \times 0.62)/(7.9 \times 10^{-7} \times 2.6 \times 10^{-3})$	**C1**	
			$F = 660$ N	**A1**	**[2]**

(iii) *either* stress when cold = 660/(7.9 x 10^{-7}) = 840 MPa

or tension at uts = 198 N **M1**

either this is greater than the ultimate tensile stress

or tension at uts is less then tension in **(ii)** **A1**

the wire will snap **A1 [3]**

(Allow possibility for the two 'A' marks to be scored as long as some quantitative answer – even if incorrect – has been given for the 'M' mark)